

CANCER SCREENING

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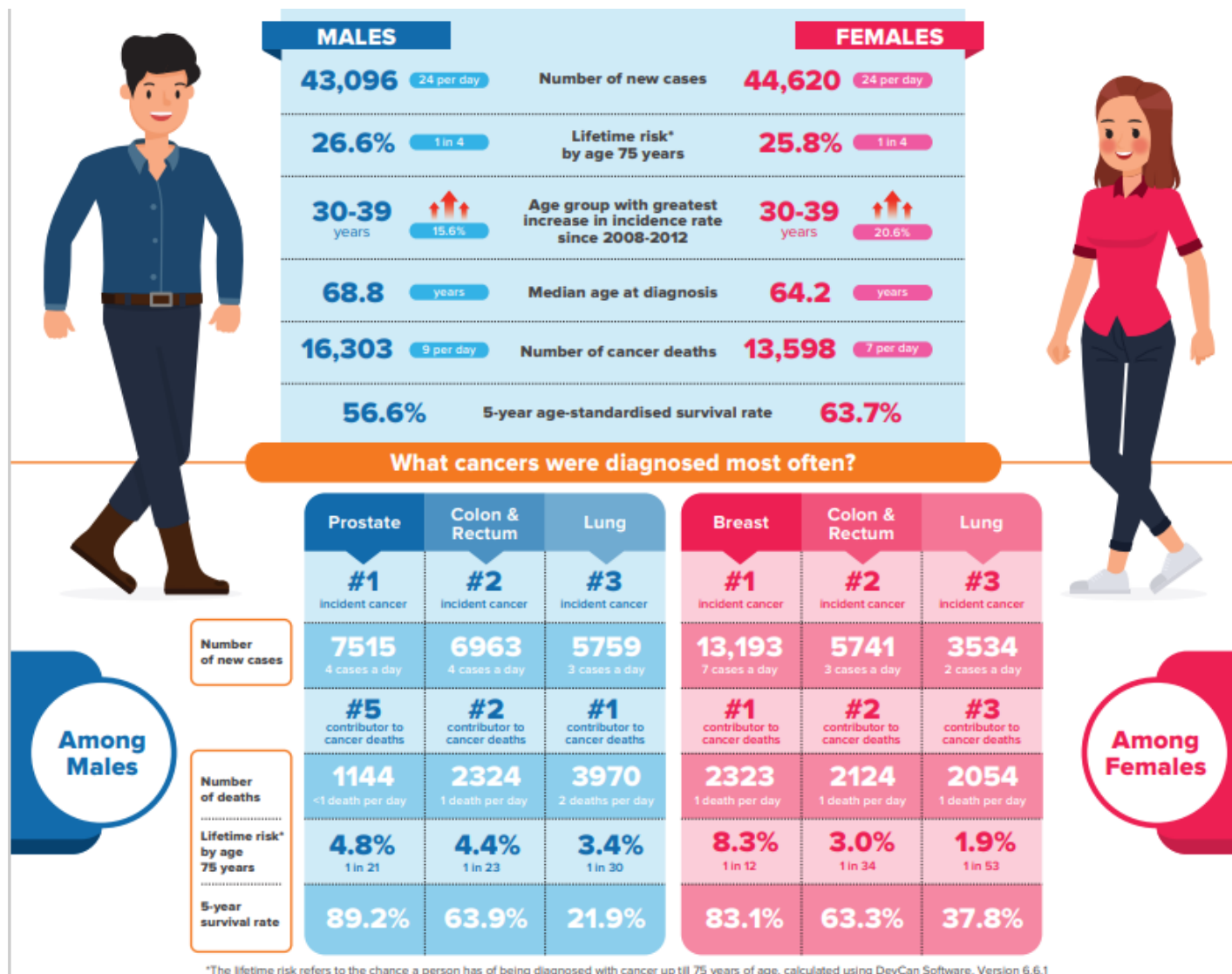
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INTRODUCTION

Importance

Cancer is currently the leading cause of death in Singapore, accounting for 24.6% of deaths in 2023.



Epidemiology

The Singapore Cancer Registry Annual Report published by the National Registry of Diseases Office includes trends for incidence, mortality and survival of cancer since 1968, with focus on most common cancers for the past five years.

Prevention

Lifestyle and behavioural factors such as obesity, physical inactivity, and smoking increase an individual's risk of developing cancer. As such, it is important to encourage healthful behaviour in individuals to minimise the impact of these risk factors on health.

SCREENING

Principles of Screening

Health screening is conducted to facilitate early diagnosis of diseases that have yet to manifest (**asymptomatic**), so that treatment and intervention can be instituted promptly to achieve good health outcomes.

Locally, a report by the screening test review committee has been published in March 2019. The committee has reviewed the appropriateness of specific screening tests based on scientific evidence of their effectiveness and the potential benefits and harms. The report tiers its recommendations by placing the screening tests into 3 categories.

Category	Definition
1. <i>Population-level screening</i>	There is good robust evidence that the screening test is both clinically effective and cost effective for use to screen the population (for the specified age range)
2. <i>Individual-level decision</i>	The net benefit does not outweigh the risk in general populations, but the screening may be useful for high-risk populations. OR There is some evidence that the screening test is effective but cost effectiveness has not been evaluated or the cost effectiveness ratio is unfavourable.
3. Not recommended	There is insufficient evidence to make a decision regarding the usefulness of the test. OR There is good evidence that the screening test is not effective, or that the net harm outweighs benefits.

This framework is not meant to replace the clinical judgment of physicians as doctors would still need to assess the suitability of specific screening tests for their patients.

SCREENING FOR COLORECTAL CANCER

How to Screen?

1. Stool based tests
 - a. Faecal Immunochemical Test (FIT)
 - b. Guaiac Faecal Occult Blood Test (FOBT) are no longer used for asymptomatic screening due to its low sensitivity and specificity (confounded by blood loss proximal to colon and not affected by diet)
 - c. Stool DNA test (not available in NUP)
2. Direct visualisation tests / Imaging
 - a. Colonoscopy
 - b. Flexible sigmoidoscopy (better if paired with FIT)
 - c. CT colonography

Note: Carcinoembryonic antigen (CEA) is NOT recommended for screening

Colorectal Cancer Screening in NUP

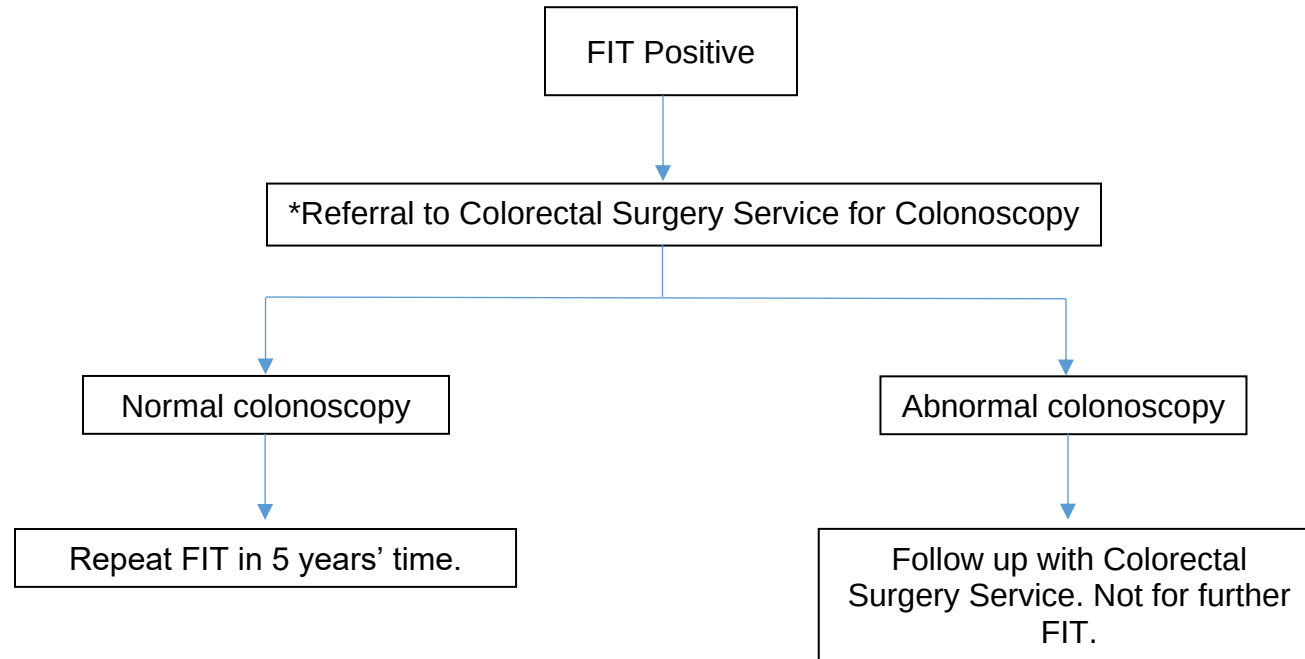
1. Colonoscopy: refer open access colonoscopy (if fulfils criteria listed on open access hardcopy form) or refer to SOC routine for screening colonoscopy
2. FIT[#]: (Colorectal Screening (via FIT) and Follow-Up Workflow)
 - a. HSG enrollee: send to Care Coordinator (CC) for FIT test, **CCs to check**
 - i. **EISS eligibility**
 - ii. **PR Bleeding – If yes, route patient to Dr for assessment**
 - b. Non-HSG enrollee: Provider to order “FECAL IMMUNOCHEMICAL TEST (FIT) PANEL”

[#] For patients at increased risk of colorectal cancer as below (full table on risk stratification and screening tools on page 7), CCs will refer to clinician for colonoscopy discussion. FIT to be done only if patient declines colonoscopy:

- Family history (first degree relative) of colorectal malignancy
- Medical history of colorectal malignancy
- Medical history of colorectal polyp
- Medical history of inflammatory bowel disease
- (Female only) Medical history of ovarian/endometrial cancer

RISK GROUP	SCREENING TOOL	ONSET (Age, Year)	FREQUENCY
A. Average Risk Asymptomatic or family history limited to non-first degree relatives	Faecal Immunochemical testing (FIT)	50	Annually
	Colonoscopy	50	Every 5-10 years
	CT Colonography	50	Every 5 years
B. Increased risk			
1. Colorectal cancer in first degree relative aged 60 years or younger or two or more first degree relatives	Colonoscopy	10 years prior to youngest case in the family or by age 40 years, whichever is earlier	Every 5 years
2. Colorectal cancer in first degree relative over the age of 60 years	Colonoscopy	10 years prior to youngest case in the family or by age 50 years, whichever is earlier	Every 5-10 years
3. Personal history of colorectal polyps	Colonoscopy	1-3 years after polypectomy in the presence of high-risk features (> 1cm, multiple, villous architecture); otherwise 3-5 years after polypectomy for low risk polyps (solitary, tubular architecture)	-
4. Personal history of colorectal malignancy	Colonoscopy	One year after resection	Every 1-3 years
5. Personal history of ovarian or endometrial cancer	Colonoscopy	One year after resection	
C. High Risk			
1. Family history of familial adenomatous polyposis	Flexible sigmoidoscopy (switch to colonoscopy if adenomas identified); consider referral for cancer genetic risk assessment	10 to 12 years (from puberty)	Annually
2. Family history of hereditary non-polyposis colorectal cancer (Lynch syndrome)	Colonoscopy; consider referral for cancer genetic risk assessment	20 - 25 years	Every 1-2 years
3. Inflammatory bowel disease a) left-sided colitis b) pan-colitis	Colonoscopy Colonoscopy	From 15 th year of diagnosis onwards From 8 th year of diagnosis onwards	Every 1-2 years Every 1-2 years

Management of Colorectal Cancer Screening



**Administrative details for follow-up can be found here: [Colorectal Screening and Follow-Up Workflow](#)*

SCREENING FOR BREAST CANCER

Who and How to Screen?

1. Mammogram
 - 40 – 49 years old: Shared decision making with patient after discussion of potential benefits, limitations, and harms of screening (e.g. Higher false positive rates, false negative results that lead to false reassurance), as well as the women's preferences and breast cancer risk profile. If mammogram is performed, it should be done annually.
 - 50 – 69 years old: Mammogram every 2 years
 - > 69 years old: Individualised decision considering benefits and risks of mammogram and estimated life expectancy. If screening is performed, it is to be done 2-yearly
2. Breast self-examination (a week after menses) is not used for screening. BSE can be encouraged for women from the age of 30 to improve women's awareness of their own breasts and breast cancer.
3. Breast MRI should not be used for screening of women who are at normal risk of developing breast cancer. It may be used as an adjunct to mammogram for high-risk groups, or for women with diffuse breast injection augmentation.
4. Ultrasound breast, tumour markers (CEA, CA15-3) and clinical breast examination are not routinely recommended for screening.

Special Populations

1. Refer to breast clinic for breast cancer screening for the following high-risk groups:
 - Received radiation treatment to the chest, such as for Hodgkin disease
 - Women with gene mutations conferring high risk of breast cancer, or with first degree relatives with these gene mutations
 - Strong family history of breast cancer but no proven mutation: Screening is recommended to commence as early as 5 - 10 years prior the age of onset in the youngest family member with breast cancer, but not earlier than age 25 to 30 years old
2. Previous breast cancer or pre-malignant conditions
 - Women with prior breast cancer or pre-malignant conditions should receive annual screening mammography of the remnant and contralateral breasts. This can be done under Breast Screen Singapore after patient is discharged to primary care under the breast cancer survivorship programme.
 - Patients with germline mutations that predispose to increased risk of breast cancers should be followed up at breast clinic.

To Arrange for a Mammogram at NUP

Patients can self-book at:

- a) Call 6370 6556
- b) NUHS or HealthHub App
- c) <https://for.sg/booknuhsdmammogram>



Mammogram services are located at the following polyclinics:

- a) Bukit Batok Polyclinic
- b) **Bukit Panjang Polyclinic**
- c) Choa Chu Kang Polyclinic
- d) Clementi Polyclinic
- e) Pioneer Polyclinic

Management of Abnormal Mammograms (Report Available on NEHR)

Patients should automatically receive a follow-up referral from Health Promotion Board to one of the following approved Assessment centres:

Medical Institution	Additional Details for Referral
Changi General Hospital (CGH)	Please call: 8127 7900
Khoo Teck Puat Hospital (KTPH)	Please call: 6602 1665
National Cancer Centre (NCC)	Please call: 6436 8415
National University Hospital (NUH)	Please call: 6772 2263
Tan Tock Seng Hospital (TTSH)	Please call: 6357 8177
Sengkang General Hospital (SKGH)	Please call: 6930 3220/6930 3990

SCREENING FOR CERVICAL CANCER

Who and How to Screen?

All females who have ever had sexual intercourse[%] aged

- 25 – 29 years: cervical cytology (pap smear), once every 3 years
- 30 – 69 years: Human Papilloma Virus (HPV) DNA test, every 5 years (recommended under National Cervical Cancer Screening Programme)

% Women who have never had sexual intercourse need not have screening. NUP does not do screening for patients with no previous sexual intercourse. Ultrasound and CT pelvis is not recommended as screening tests.

When to Stop Screening

1. A woman can be discharged from screening at 69 years of age if she has:
 - 3 consecutive negative cervical cytology tests
 - or 2 consecutive negative HPV tests in the last 10 years, with the most recent test occurring within the last 5 years
2. For women who had history of CIN2, CIN3 or AIS, routine screening should continue for at least 20 years, even if it extends beyond 69 years of age.

Special Populations

1. Immunocompromised women (HIV positive, undergone solid organ transplant, or clinical conditions requiring 2 or more immunosuppressive agents):
 - Annual cervical cytology for women between 25 to 29 years old
 - 3 yearly HPV primary screening for women who are 30 years old and above:
 - Those who are tested with any high-risk HPV strains should be sent for colposcopy instead of doing a cytology triage
 - Lifetime screening

2. Women who have undergone hysterectomy:

Subtotal hysterectomy	Routine cervical cancer screening
Hysterectomy for benign disease, and no known cervical cancer precursors / cancer	Stop screening
Hysterectomy for unknown histology	Do 1 baseline vault smear, stop screening if negative
Immunosuppressed (HIV positive, undergone solid organ transplant, or clinical conditions requiring 2 or more immunosuppressive agents)	Vault smears yearly
Past history of CIN	1. If excision margin was involved or not adequately assessed histologically: follow up should be at the discretion of the gynaecologist. In general, vault smear at least yearly. 2. CIN 1/2/3 completely excised during hysterectomy: Vault smear for five years yearly, then 2-yearly subsequently
Past history of invasive gynaecological cancer, or previously treated for vaginal intra-epithelial neoplasia	Follow up with gynaecologist.

Note: Vault smears are not included in the CSS programme.

Cervical Cancer Screening in NUP

Order "TCU NUR Cervical Cancer screening (First Visit)" on NGEMR, patient to book appointment at kiosks as per usual appointments.

Management of Abnormal Cervical Cancer Screening Results

Refer to [Cervical Screening Guidelines 2019](#) ; If inflammatory cells reported in < 30-years-old, treat only if symptomatic; no follow-up is required if asymptomatic

HPV Vaccination

Offer HPV vaccination for females aged 9 - 26 years to reduce risk of cervical cancer

School-based programme since April 2019	Settings outside of school-based programme
Dose 1: HPV2-valent at 12-13 years (Secondary 1) Dose 2: HPV2 at 13-14 years (Secondary 2) Dose 3 is only recommended if dose 1 was given at 15 years of age or older.	Outside of school-based programme, HPV vaccination continues to be recommended in NCIS and NAIS as a catch-up with age-appropriate doses for females up to and through 26 years of age

HPV testing should NOT be used for screening before deciding on HPV vaccination.

SCREENING FOR ENDOMETRIAL CANCER

Who to Screen?

1. Women with hereditary non-polyposis colorectal cancer (HNPCC) or Lynch Syndrome may consider annual screening starting between the ages of 30 and 35.
2. Routine screening of the following is not recommended:
 - a) Women with *average risk* (no risk factors), or
 - b) Women with *increased risk* (e.g. obesity, diabetes, hypertension, nulliparity, infertility, ovulation failure, late menopause, tamoxifen therapy or history of unopposed oestrogen therapy).

Women with endometrial cancer tend to present with symptoms at an early favourable stage. Transvaginal Ultrasound screening for endometrial cancer has good sensitivity before symptoms appear, however there is limited mortality benefit.

Although endometrial cancer screening is not recommended for the general population, early evaluation of postmenopausal bleeding with referral to the gynaecologist is important for early detection.

SCREENING FOR OVARIAN CANCER

Who to Screen?

1. There is insufficient supporting evidence for the routine screening of asymptomatic women at increased risk of ovarian cancer. Refer such women to tertiary centres for assessment.
2. High Risk Group: Known BRCA-carriers should be on follow up with an oncologist. Transvaginal pelvic ultrasound as screening for ovarian cancer may be considered. However, it can result in unnecessary surgeries with no clear mortality benefits. The potential benefits and limitations should be discussed with the patient for informed decision-making.
3. Present evidence does NOT support routine screening with serum markers (e.g. CA 125) and/or ultrasound in this group of women as it is ineffective and tends to lead to unnecessary interventions, as well as risks to health and well-being.

SCREENING FOR PROSTATE CANCER

Who to Screen?

1. Current evidence does not support population-based screening.
2. May offer screening to men aged 50 - 70 years, with an estimated further life expectancy of more than 10 years. Discuss potential benefits and risks before coming to a shared individualised decision. Can consider using SmartPhrase .INPSA to document discussion within NGEMR.
3. High risk groups, such as men with a strong family history of prostate cancer, i.e. one or more first-degree relatives (father, brother) diagnosed before age 65 years, may be offered screening 5 to 10 years younger than the youngest prostate cancer in the family.

How to Screen?

1. Serum Prostate Specific Antigen (PSA) is recommended.
 - a) PSA > 4 ng/ml requires further follow-up with a urologist to exclude prostate cancer.
 - b) Benign causes of elevated PSA include BPH, prostatitis, recent prostate biopsy / TURP / cystoscopy, ejaculation, urinary retention, perineal trauma or prostatic infarction. Digital rectal examination (DRE) does not cause a clinically significant rise in PSA.
 - c) PSA is lowered by 5 alpha reductase inhibitors (5ARI) by about 50% after 6-12 months. A rise in PSA for a patient on 5ARI should be considered for urology referral.
2. DRE is not recommended for screening, although it may be used to evaluate an elevated PSA, or as an adjunct along with PSA testing. If performed, any prostate nodules, induration or asymmetry would warrant a referral to urology.

How Frequently to Screen?

1. A routine screening interval of two years or more may be preferred over annual screening in asymptomatic men who have participated in shared decision making and decided on screening.
2. For patients who were previously on follow up with urology, the interval PSA screening should be continued as per previous decision made by discharging urologist.

SCREENING FOR GASTRIC CANCER

1. Current evidence does not support population-based screening.
2. High risk groups:
 - Individuals with HNPCC or Lynch Syndrome may benefit from screening with oesophago-gastro duodenoscopy (OGD), starting from age 30 - 35 years.
 - Individual-level screening, refer to GASTROClear section under Gastro SAG for eligibility and workflow information.

SCREENING FOR LIVER CANCER

Who to Screen?

1. No need to screen general population.
2. Offer screening to high-risk groups: Hepatitis B carrier or individuals with cirrhosis.

How to Screen?

1. High Risk patients: 6-monthly alpha-fetoprotein and 6 – 12 monthly ultrasound HBS (refer to Hepatitis B CPG for more details).
2. Liver function test is not recommended as a screening test for liver cancer.

SCREENING FOR LUNG CANCER

1. Current evidence does not support population-based screening.
2. Annual Low-dose CT (LDCT) screening may be offered in the following high-risk groups after discussion of benefits and risks (individual level decision).
 - a. Individuals aged between 55 to 74 who have smoked ≥ 30 pack years and are continuing to smoke
 - b. Individuals aged between 55 to 74 who have smoked ≥ 30 pack years but quit <15 years ago

This is based on the findings of the National Lung Screening Trial (NLST) in US which demonstrated a reduction in lung cancer mortality and all-cause mortality with LDCT screening in high-risk populations. However, screening should be balanced with the unnecessary exposure to radiation, financial costs of CT screening, and a high rate of false positive findings that may result in unnecessary further evaluation and its accompanying risks.

In Singapore, there has been an increased incidence of lung adenocarcinoma. This reflects the Asia-Oacific trend of higher rates of adenocarcinoma, particularly in women and in non-smokers. Further evidence from local population research is ongoing for local lung cancer screening guidelines.

3. If patient is agreeable for screening, refer Respiratory medicine (routine) and indicate under reason for referral: Others (High risk group for discussion for Lung Ca screening)
4. CXR or tumour markers for lung cancer are not recommended as screening tools.

SCREENING FOR NASOPHARYNGEAL CARCINOMA

Who to Screen?

1. No need to screen general population.
2. May offer screening to high-risk groups: Individuals with a first degree relative (parent, sibling) with NPC.

How to Screen?

1. Anti-EBV EA IgA and nasoendoscopy. Refer to ENT specialist for screening.

GUIDELINES FOR GENETIC TESTING REFERRAL TO MEDICAL ONCOLOGY (CANCER GENETICS CLINIC)

1. Hereditary breast and ovarian cancer syndrome:
 - a. Personal or family history of
 - breast cancer diagnosed < 40 years of age
 - male breast cancer, any age
 - epithelial ovarian cancer, any age
 - triple negative breast cancer diagnosed < 50 years of age
 - ≥ 2 breast cancers, at least one aged < 50 years of age
 - both breast and epithelial ovarian cancers
 - Two or more breast/ovarian cancers in the same patient
 - b. Individual from a family with a known BRCA1/2 mutation or other rare gene mutations in the family
 - *BRCA1/2 mutation carriers are also at risk for pancreatic and prostate cancer and clinicians should additionally ask for these cancers in taking the family history.*
 - *Physicians should consider referring families with breast cancer and young onset pancreatic or prostate cancer diagnosed before age 50.*
2. Lynch Syndrome (LS)
 - a. Patient with CRC:
 - i. diagnosed <50;
 - ii. ≥ 1 primary CRC or other LS-related tumours[^]
 - iii. diagnosed at any age,
 - and who has at least 1 first-degree relative diagnosed with CRC or LS-related tumour diagnosed <50 years;
 - and who has 2 or more first- or second-degree relatives with CRC or LS-related tumour[^] at any age.
 - b. Meets Amsterdam criteria (At least 3 family members affected with cancer, spanning 2 generations, with at least 1 affected family member diagnosed below age 50 years; the 3 affected family members have to be first-degree relatives of each other.
 - c. Endometrial cancer < 50 years
 - d. Known LS mutation in family
3. Familial Adenomatosis Polyposis (FAP)

[^] LS-related tumours: endometrium, stomach, pancreas, small intestine, ovary, kidney, brain, ureters, or bile duct

^{**} Family history would be in reference to first- and second-degree relatives.

REFERENCES

1. Report of the Screening Test Review Committee, Academy of Medicine, Singapore. March 2019
2. Management Guidelines for Cervical Screening & Preinvasive Disease of the Cervix. February 2019
3. MOH Circular No. 180/2020 Changes to the National Childhood Immunisation Schedule. 13 July 2020
4. Healthier SG Care Protocols Cancer Screening (primarycarepages.sg) September 2024
5. Lim TSC, Ismail-Pratt I, Goh LH. Cervical cancer screening and vaccination: understanding the latest guidelines. Singapore Med J. 2022 Mar;63(3):125-129. doi: 10.11622/smedj.2022045. PMID: 35509250; PMCID: PMC9251248.
6. Singapore Cancer Network (SCAN) Cancer Genetics Workgroup. Singapore Cancer Network (SCAN) Guidelines for Referral for Genetic Evaluation of Common Hereditary Cancer Syndromes. Ann Acad Med Singapore. 2015 Oct;44(10):492-510. PMID: 26763067.